# NPC – UNIT 4 ASSIGNMENT

Configure DHCP and Set Up a Network Using Cisco Packet Tracer

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| PRN | 2125UCSF1071 |
| Subject | Network Protocols and Communication (NPC) |
| Software | Cisco Packet Tracer 9.0.1 |
| Assignment Unit | Unit 4 – DHCP Configuration |
| GitHub Repository | github.com/rutujabhalerao19-r/NPC |

Submitted as a step-by-step practical report with screenshots captured during the configuration and testing process.

# 1. Aim

To create a small LAN in Cisco Packet Tracer, configure a server with a static IPv4 address, enable DHCP, assign IP configuration automatically to PCs, and verify connectivity using ping.

# 2. Network Topology

The topology contains three PCs, one 2960-24TT switch, and one server. Each end device is connected to the switch using Copper Straight-Through Ethernet cables.

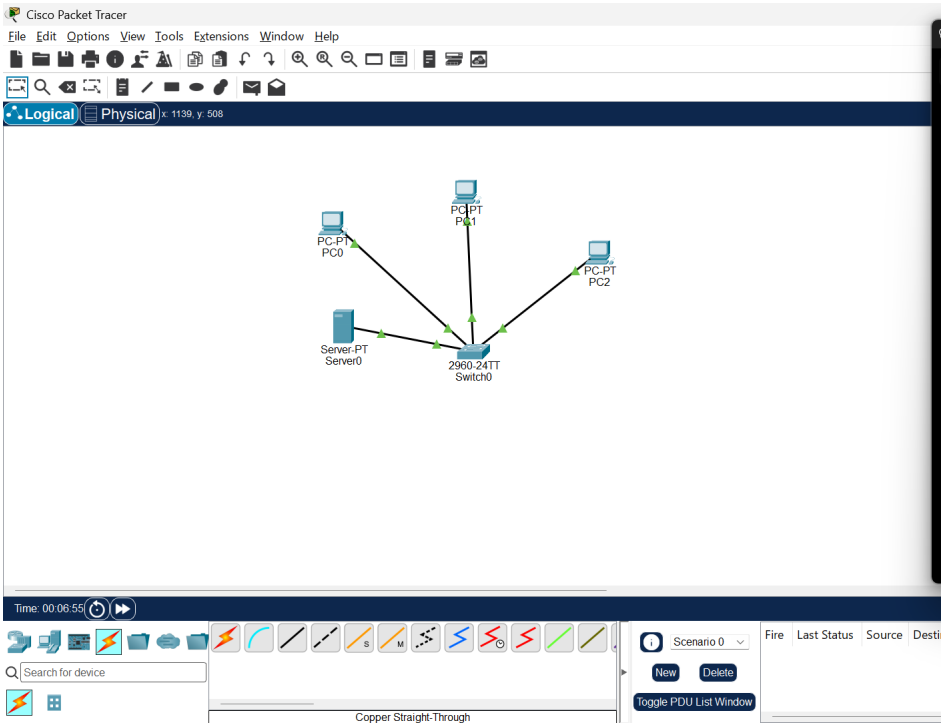



*Figure 1. Network topology with PC0, PC1, PC2, Server0, and Switch0.*

# 3. IP Addressing and DHCP Configuration

Server0 was configured with a static IPv4 address so that clients can communicate with it reliably. The DHCP service was enabled and a pool named NPC_POOL was configured.

| Device / Setting | Configured value |
|---|---|
| Server0 IPv4 address | 192.168.1.2 |
| Subnet mask | 255.255.255.0 |
| DHCP pool name | NPC_POOL |
| DHCP pool start address | 192.168.1.10 |
| DHCP pool subnet mask | 255.255.255.0 |
| Maximum number of users | 50 |
| Default gateway / DNS | 0.0.0.0 (not configured for this local LAN test) |

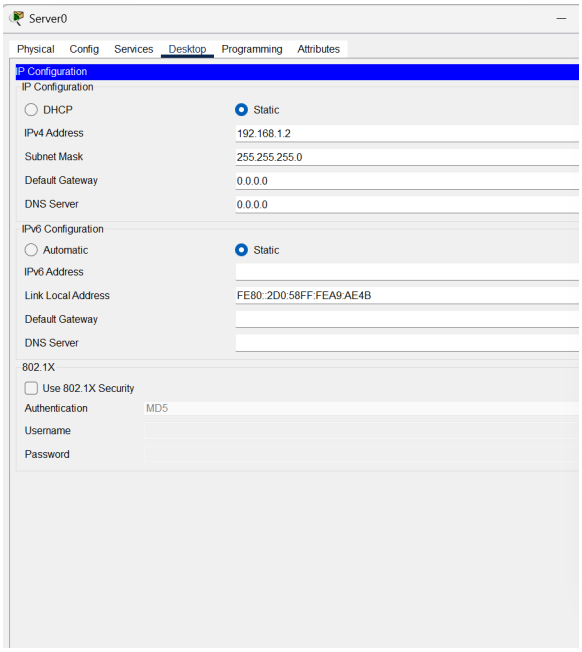



*Figure 2. Server0 static IP configuration.*

# 4. DHCP Service Setup

In Server0, Services → DHCP was opened. DHCP service was switched On, and the NPC_POOL entry was configured with the start IP address 192.168.1.10 and subnet mask 255.255.255.0.

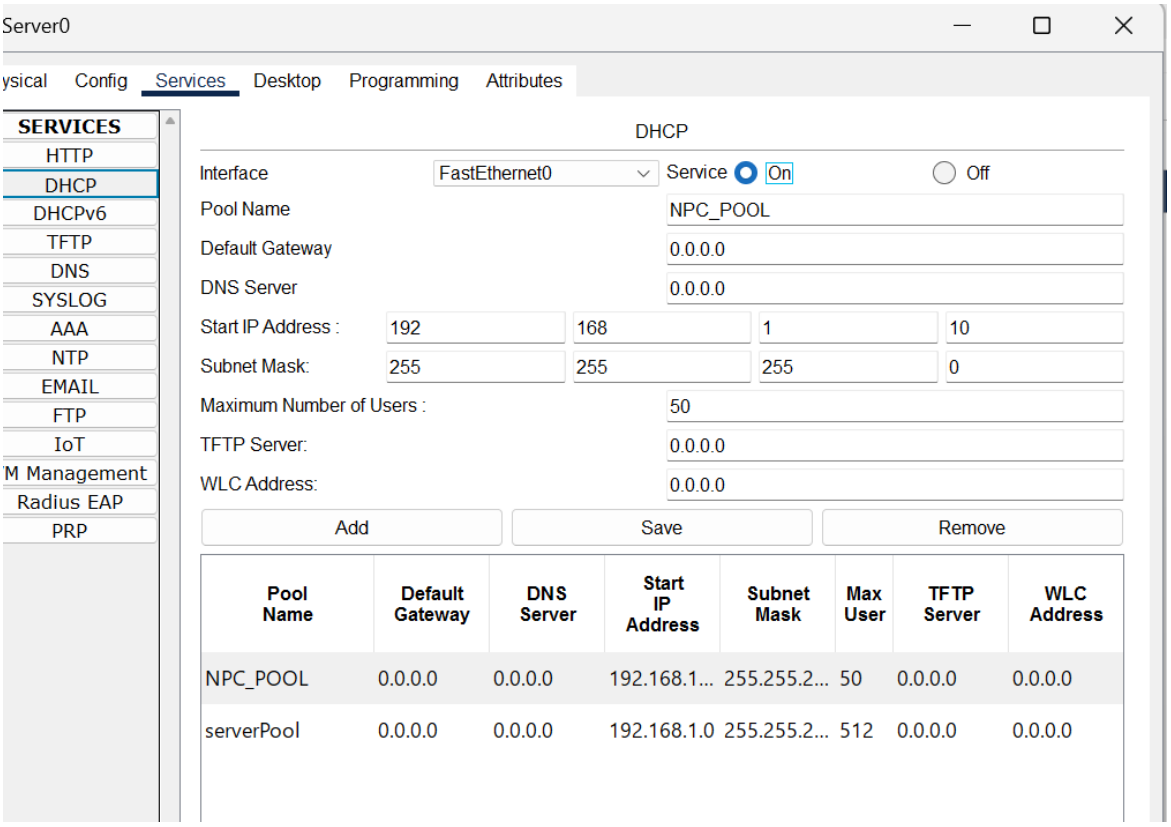



*Figure 3. DHCP service and NPC_POOL settings.*

# 5. Automatic IP Configuration on PCs

On each PC, Desktop → IP Configuration was opened and DHCP was selected. The screenshots show successful DHCP requests and the IPv4 addresses received by the PCs.

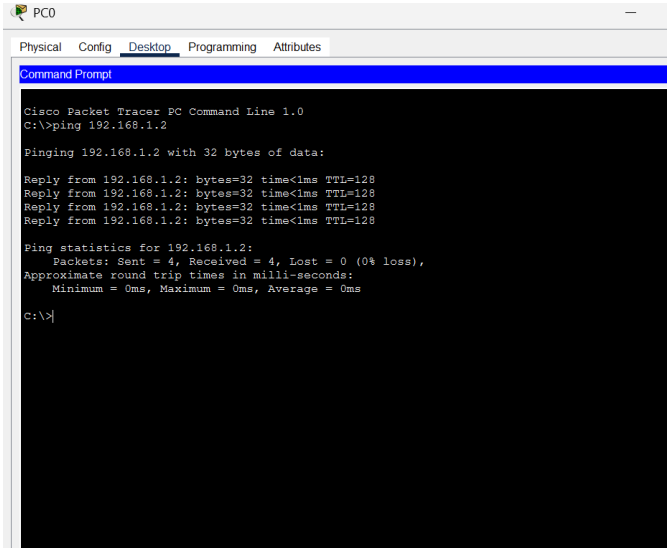



*Figure 4. PC0 IP configuration (DHCP successful; IPv4 address shown as 192.168.1.1).*

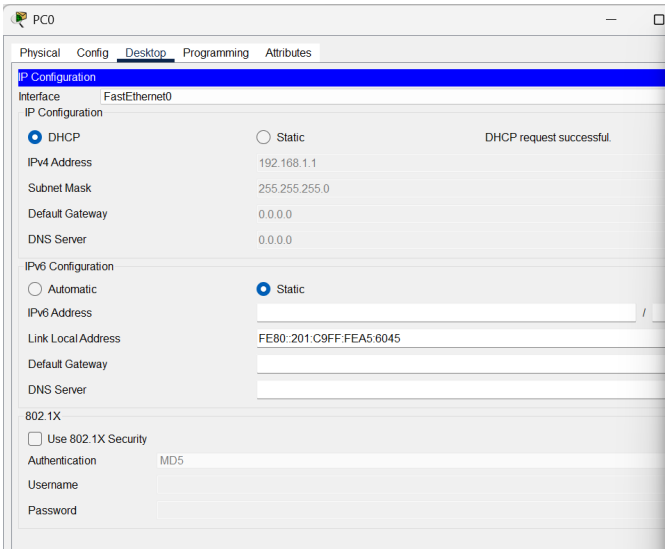



*Figure 5. PC1 IP configuration (DHCP successful; IPv4 address shown as 192.168.1.3).*

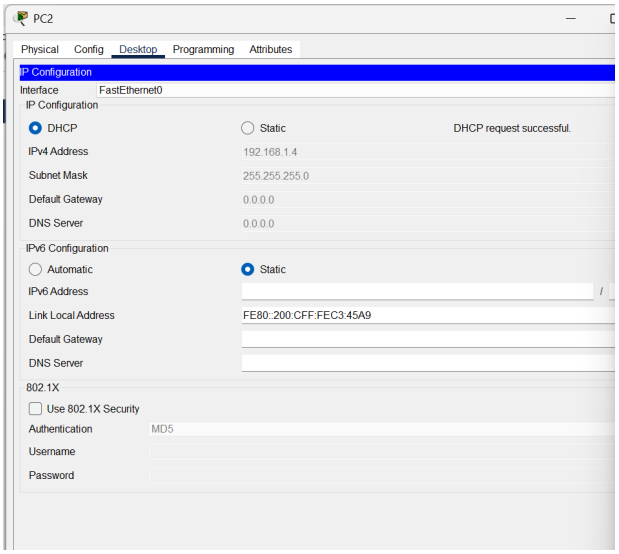



*Figure 6. PC2 IP configuration (DHCP successful; IPv4 address shown as 192.168.1.4).*

## 6. Connectivity Testing

Connectivity was verified from PC0 using the ping command. The tests to Server0 (192.168.1.2), PC1 (192.168.1.3), and PC2 (192.168.1.4) returned replies. The displayed statistics show 4 packets sent, 4 received, and 0% packet loss for each test.

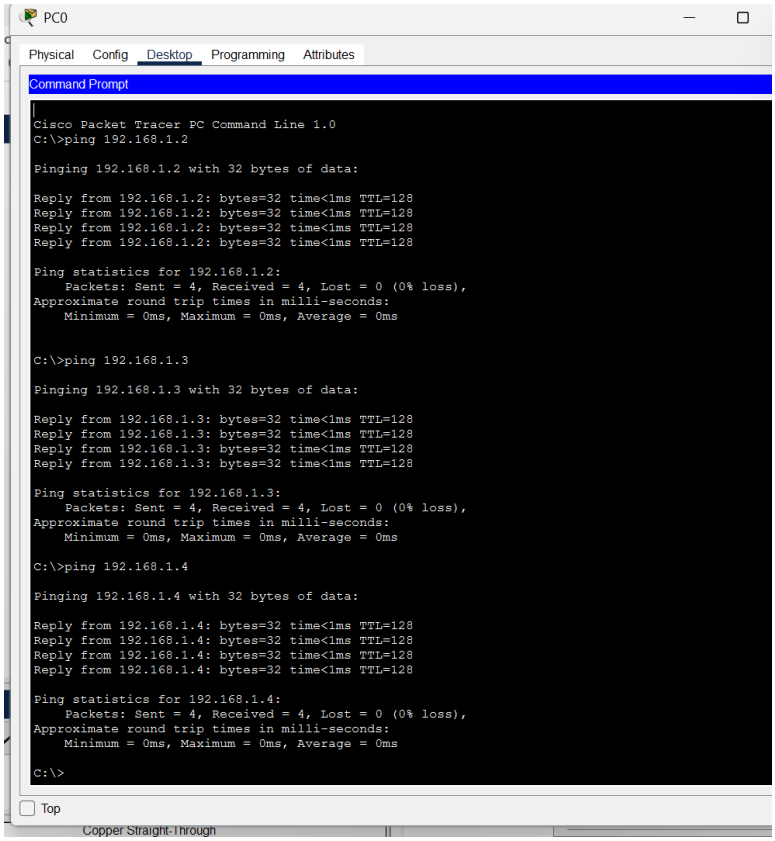



*Figure 7. Successful ping tests from PC0 to Server0, PC1, and PC2 (ChatGPT window excluded).*

# 7. Results and Conclusion

The LAN topology was created successfully in Cisco Packet Tracer. Server0 was assigned a static IP address, the DHCP service was enabled, and the PCs received IPv4 configuration automatically. Ping tests confirmed connectivity between PC0, Server0, PC1, and PC2.

**Configuration note:** The captured PC screenshots show addresses 192.168.1.1, 192.168.1.3, and 192.168.1.4, although the NPC_POOL was configured to start at 192.168.1.10. This indicates that the default serverPool or an earlier DHCP lease may have supplied the displayed client addresses. The network connectivity tests succeeded; the DHCP pool allocation should be reviewed if the instructor requires the clients to receive addresses specifically from .10 onward.